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Unconditionally maximum principle preserving scheme for the Allen–Cahn equation with a logarithmic free energy

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ABSTRACT

We present a novel computational scheme that unconditionally satisfies the maximum principle to solve the Allen–Cahn (AC) equation with a logarithmic Flory–Huggins potential. The proposed scheme uses an operator splitting approach integrated with a frozen coefficient technique without using any stabilization term. Due to the use of the frozen coefficient method, we can apply a closed-form solution to the highly nonlinear term. This combination allows the development of a scheme that rigorously maintains the maximum principle throughout the computation. We provide a detailed analytical proof of the discrete maximum principle for the proposed scheme to ensure its theoretical robustness. To validate the high performance, we conduct computational experiments, which confirm the unconditional stability and demonstrate the method's effectiveness in preserving the maximum principle. These numerical results highlight the proposed scheme's potential as a reliable tool for accurately solving the AC equation.

1. Introduction

We present an unconditionally maximum principle satisfying methodology for the Allen–Cahn (AC) equation with a logarithmic energy:

$$\frac{\partial \phi}{\partial t} = -\frac{F'(\phi)}{\epsilon^2} + \Delta\phi = -\frac{\theta}{2\epsilon^2}\ln\left(\frac{1+\phi}{1-\phi}\right) + \frac{\theta_c}{\epsilon^2}\phi + \Delta\phi, \tag{1}$$

where $\phi = \phi(x, y, t)$, ϵ is the gradient energy coefficient, and Ω is a domain in $\mathbb{R}^2$. The AC equation (1) is obtained from the following functional as the gradient flow [1]:

$$\mathcal{E}(\phi) := \int_\Omega \left(\frac{F(\phi)}{\epsilon^2} + \frac{1}{2}|\nabla\phi|^2\right) d\mathbf{x}, \tag{2}$$

where

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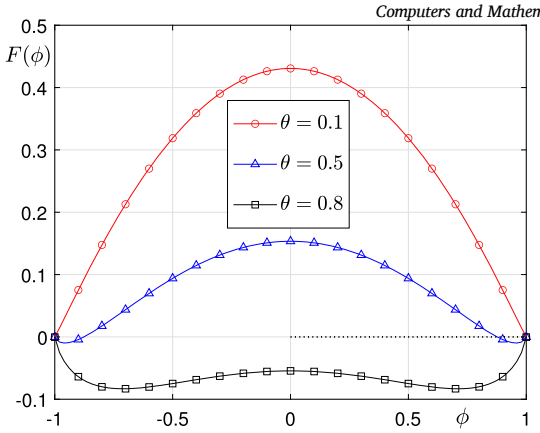


Fig. 1. Logarithmic free energy $F(\phi)$, for various θ values. Here, $\theta_c = 1$ is used.

$$F(\phi) = \frac{\theta}{2}\left[(1+\phi)\ln\left(\frac{1+\phi}{2}\right) + (1-\phi)\ln\left(\frac{1-\phi}{2}\right)\right] + \frac{\theta_c}{2}(1-\phi^2). \tag{3}$$

Here, θ and θ_c denote the absolute temperature and the critical temperature, respectively. Fig. 1 shows a graph of the logarithmic free energy $F(\phi)$ with the critical value $\theta_c = 1$ fixed constant: $\theta = 0.1$ (red circles), $\theta = 0.5$ (blue triangles), and $\theta = 0.8$ (black squares).

The AC equation is extensively used for modeling phase transitions and interfacial dynamics [2]. The AC equation, along with its associated equations, is broadly applied across numerous research disciplines, including topology optimization [3], image processing [4], crystal growth [5], and fluid simulation [6]. The logarithmic free energy potential is important in the AC equation for accurately modeling phase separation and interfacial dynamics. Unlike polynomial potentials, the logarithmic form, derived from the Flory–Huggins theory, effectively captures the singular behavior near critical concentrations, which ensures that the phase field variable remains within physically meaningful bounds [7].

This characteristic is essential for simulating realistic phase transitions in binary systems. Recent studies have focused on developing efficient numerical schemes to handle the challenges posed by the strong nonlinearity and singularity of the logarithmic potential. Wang et al. [8] introduced a stabilized energy factorization algorithm for solving the AC equation. This scheme is a semi-implicit, linear method that satisfies the maximum principle and energy dissipation law, with rigorous proofs and numerical validations. Xiao and Feng [9] proposed an operator splitting finite element method (FEM) for solving the AC equations. Kim and Lee [7] considered a numerical algorithm for the AC equation with logarithmic and polynomial potentials, and analyzed their impact on energy dissipation and morphological dynamics. It highlights discrepancies between logarithmic potentials and their polynomial approximations. Wang et al. [10] developed a two-grid FEM for the AC equation by combining a coarse-grid nonlinear solver and a fine-grid linearized solver. Numerical examples confirmed the method's energy stability, optimal convergence in the H^1 norm, and significant computational efficiency. Wu et al. [11] proposed a second-order Strang splitting method with an exponential integrating factor, which overcomes challenges posed by the logarithmic term's singularity, as demonstrated through rigorous analysis and validated by 2D and 3D simulations. Lv et al. [12] proposed a reduced-order model of the Cahn–Hilliard (CH) equation with a logarithmic energy potential based on deep learning, which offers computational savings. Chen et al. [13] proposed finite difference schemes for the CH equation which offer first and second-order temporal accuracy. The schemes ensure positivity, energy stability, and optimal convergence rates, using the singularity of the logarithmic term to prevent solutions from reaching singular values, as validated by theoretical analysis and numerical experiments. Zhang et al. [14] proposed delay-free stabilization algorithms for the AC gradient flow system to resolve the computational challenges caused by intractable time delays in traditional stabilization techniques. By using Runge–Kutta integrators and a relaxation technique, the proposed methods ensure unconditional energy stability, mesh-dependent maximum principle preservation, and improved temporal accuracy, as validated by numerical experiments. Conti et al. [15] investigated a perturbed CH equation with logarithmic nonlinear terms, and proved the existence, uniqueness, regularity, and separation properties of solutions. Furthermore, the study demonstrates convergence to the CH equation over finite time intervals. In addition, numerical methods for equations related to or similar to the AC equation have been extensively studied, including the scalar auxiliary variable method [16,17], Lagrange multiplier method, convex splitting method, and operator splitting method [18]. In this work, the operator splitting scheme is specifically adopted in the construction of the numerical algorithm due to its simplicity and efficiency.

The primary purpose of this paper is to propose an unconditionally maximum principle-satisfying algorithm for solving the AC equation with a logarithmic potential using a frozen coefficient technique.

This paper is organized as follows. Section 2 presents a new numerical scheme. Section 3 provides computational tests that validate the proposed method's unconditional stability. Section 4 concludes this paper.

2. Computational method

Let $\Omega = (0,1)\times(0,1)$ and $\Omega_h = \{(x_i, y_j) | x_i = h(i-0.5),\ 1 \le i \le N_x \ \text{ and } \ y_j = h(j-0.5),\ 1 \le j \le N_y\}$. While a uniform step size h is used for simplicity, the method can be extended to nonuniform grid sizes to enhance efficiency, as illustrated in [19]. The operator

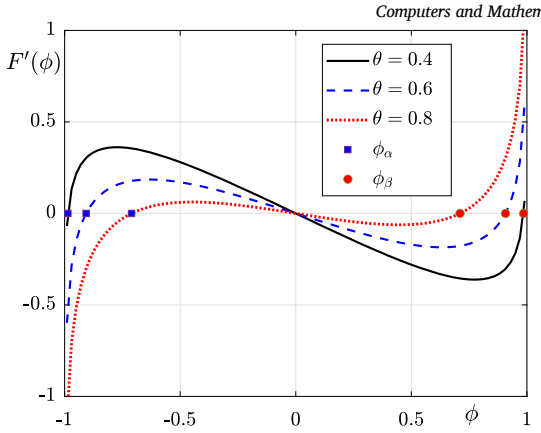


Fig. 2. Derivatives of the logarithmic free energy, $F'(\phi)$, for various θ values with ϕ_α and ϕ_β, which satisfy $F'(\phi_\alpha) = 0$ and $F'(\phi_\beta) = 0$. Here, $\theta_c = 1$ is used.

splitting method is used to solve differential equations by dividing them into less complex subproblems [20,21]. Each subproblem, being easier to handle, is solved sequentially [22]. The combined solutions of subproblems provide a numerical solution to the original problem.

$$Step\ 1: \frac{\partial \phi}{\partial t} = -\frac{F'(\phi)}{\epsilon^2}. \tag{4}$$

$$Step\ 2: \frac{\partial \phi}{\partial t} = \frac{\partial^2 \phi}{\partial x^2}. \tag{5}$$

$$Step\ 3: \frac{\partial \phi}{\partial t} = \frac{\partial^2 \phi}{\partial y^2}. \tag{6}$$

We begin by considering the following nonlinear equation in $Step\ 1$:

$$\frac{\partial \phi}{\partial t} = -\frac{\theta}{2\epsilon^2} \ln\left(\frac{1+\phi}{1-\phi}\right) + \frac{\theta_c}{\epsilon^2}\phi, \tag{7}$$

with ϕ_{ij}^n. We rewrite Eq. (7) as follows:

$$\frac{\partial \phi}{\partial t} = A\phi(\phi_\alpha - \phi)(\phi - \phi_\beta), \tag{8}$$

where ϕ_α and ϕ_β are argument minima of $F(\phi)$, and

$$A = \frac{-\frac{\theta}{2\epsilon^2} \ln\left(\frac{1+\phi}{1-\phi}\right) + \frac{\theta_c}{\epsilon^2}\phi}{\phi(\phi_\alpha - \phi)(\phi - \phi_\beta)}. \tag{9}$$

Here, we obtain Eq. (9) by dividing the right-hand side of Eq. (7) by $\phi(\phi_\alpha - \phi)(\phi - \phi_\beta)$. For simplicity, let $\theta_c = 1$. Using Newton's method, we can compute $\phi_\alpha < 0$ and $\phi_\beta > 0$, which satisfy $F'(\phi_\alpha) = 0$ and $F'(\phi_\beta) = 0$. Due to the symmetric property, we only solve ϕ_β, and then set $\phi_\alpha = -\phi_\beta$. The following is a pseudo-code for Newton's method:

Input: Function $F'(\phi)$, its derivative $F''(\phi)$, initial guess ϕ_0, tolerance tol, maximum iterations max_iter
Output: Approximate root ϕ_β

1. Initialize $\phi_\beta \leftarrow \phi_0$
2. For $i = 1$ to max_iter do:
   (a) If $|F'(\phi_\beta)| <$ tol, then:
   Return ϕ_β *// Root found within the desired tolerance*
   (b) Else, update $\phi_\beta \leftarrow \phi_\beta - F'(\phi_\beta)/F''(\phi_\beta)$ *// Newton's update step*
3. End For

Fig. 2 displays the derivatives of the logarithmic free energy, $F'(\phi)$. The curves are plotted for three specific θ values ($\theta = 0.4, 0.6, 0.8$), represented by solid, dashed, and dotted lines, respectively. The values ϕ_α and ϕ_β, shown as blue filled squares and red filled circles, respectively, are critical points where the derivative $F'(\phi)$ equals zero ($F'(\phi_\alpha) = 0$ and $F'(\phi_\beta) = 0$). For smaller values of θ (e.g., $\theta = 0.4$), the curve exhibits distinct critical points, which implies a strong separation between phases. As θ increases (e.g., $\theta = 0.8$), the critical points ϕ_α and ϕ_β shift, and the derivative becomes less steep, which indicates a weakening phase separation.

To efficiently solve Eq. (8), we use the frozen coefficient technique [23], which simplifies the solution of differential equations with variable coefficients by keeping the coefficients constant. By doing so, the complex equation is reduced to one with constant coefficients, which is more straightforward to solve. This method enables the approximation of solutions for nonlinear systems and enhances computational efficiency. For simplicity, let us introduce $\psi(t)$ and rewrite Eq. (8) as follows:

$$\frac{d\psi(t)}{dt} = -A\psi(t)(\psi(t)+\phi_\beta)(\psi(t)-\phi_\beta), \tag{10}$$

with $\psi(0)=\psi_0$. Here, we used $\phi_\alpha = -\phi_\beta$. Rewrite the equation to separate variables:

$$\frac{1}{-\psi(\psi-\phi_\beta)(\psi+\phi_\beta)}\,d\psi = A dt.$$

Using partial fraction decomposition, we have

$$\frac{1}{\phi_\beta^2}\left(\frac{1}{\psi} - \frac{1}{2(\psi-\phi_\beta)} - \frac{1}{2(\psi+\phi_\beta)}\right) d\psi = A dt.$$

Now, integrate term by term on the left-hand side:

$$\frac{1}{\phi_\beta^2}\left(\ln|\psi| - \frac{1}{2}\ln|\psi-\phi_\beta| - \frac{1}{2}\ln|\psi+\phi_\beta|\right) = At + C, \tag{11}$$

where C is the constant of integration. This constant can be specified by setting $t=0$, which leads to

$$C = \frac{1}{\phi_\beta^2}\left(\ln|\psi_0| - \frac{1}{2}\ln|\psi_0-\phi_\beta| - \frac{1}{2}\ln|\psi_0+\phi_\beta|\right). \tag{12}$$

Next, to simplify the logarithmic terms in Eq. (11), combine them into a single expression as follows:

$$\ln\frac{|\psi|}{\sqrt{|\psi-\phi_\beta||\psi+\phi_\beta|}} = \phi_\beta^2(At+C).$$

Then, we express ψ^2 explicitly as:

$$\psi^2 = \frac{\phi_\beta^2 e^{2\phi_\beta^2(At+C)}}{1+e^{2\phi_\beta^2(At+C)}} = \frac{\phi_\beta^2\psi_0^2}{e^{-2\phi_\beta^2 At}(\phi_\beta^2-\psi_0^2)+\psi_0^2},$$

where we substituted the expression for C, Eq. (12). Finally, taking the square root provides the explicit solution:

$$\psi(t) = \frac{\phi_\beta\psi_0}{\sqrt{e^{-2\phi_\beta^2 At}(\phi_\beta^2-\psi_0^2)+\psi_0^2}}.$$

Hence, by applying the frozen coefficient technique, we have a closed-form solution:

$$\psi_{ij}^* = \frac{\phi_\beta\phi_{ij}^n}{\sqrt{e^{-2\phi_\beta^2 A_{ij}^n\Delta t}(\phi_\beta^2-(\phi_{ij}^n)^2)+(\phi_{ij}^n)^2}}. \tag{13}$$

Here,

$$A_{ij}^n = \frac{-\frac{\theta}{2\epsilon^2}\ln\left(\frac{1+\phi_{ij}^n}{1-\phi_{ij}^n}\right)+\frac{\theta_c}{\epsilon^2}\phi_{ij}^n}{\phi_{ij}^n(\phi_\beta+\phi_{ij}^n)(\phi_\beta-\phi_{ij}^n)}. \tag{14}$$

If $0<\phi_{ij}^n<\phi_\beta$, then let $g(\phi)$ be the numerator of Eq. (14), i.e.,

$$g(\phi) = -\frac{\theta}{2\epsilon^2}\ln\left(\frac{1+\phi}{1-\phi}\right)+\frac{\theta_c}{\epsilon^2}\phi. \tag{15}$$

Taking derivatives $g(\phi)$, we have

$$g'(\phi) = -\frac{\theta}{2\epsilon^2}\left(\frac{1}{1+\phi}+\frac{1}{1-\phi}\right)+\frac{\theta_c}{\epsilon^2}, \tag{16}$$

$$g''(\phi) = \frac{\theta}{2\epsilon^2}\left(\frac{1}{(1+\phi)^2}-\frac{1}{(1-\phi)^2}\right). \tag{17}$$

Therefore, $g''(\phi) < 0$ for $0 < \phi < \phi_\beta$, which implies $g(\phi)$ is a concave function for $0 < \phi < \phi_\beta$. Furthermore, $g(0) = 0$ and $g(\phi_\beta) = 0$. Hence, $g(\phi_{ij}^n) > 0$, which implies $A_{ij}^n > 0$ for $0 < \phi_{ij}^n < \phi_\beta$ because the denominator of A_{ij}^n is positive. Similarly, for $\phi_\alpha < \phi < 0$, $g''(\phi) > 0$, which implies that $g(\phi)$ is a convex function within the interval $\phi_\alpha < \phi < 0$. In addition, $g(\phi_\alpha) = 0$ and $g(0) = 0$. As a result, $g(\phi_{ij}^n) < 0$, which leads to $A_{ij}^n > 0$ for $\phi_\alpha < \phi_{ij}^n < 0$ because the denominator of A_{ij}^n is negative.

We note that, since the frozen coefficient method is used, the term A in Eq. (10) is evaluated at time level n. Consequently, constructing a temporally second-order numerical scheme that is unconditionally stable and does not require iterative procedures poses a significant challenge. Nonetheless, the development of higher-order schemes that retain unconditional stability is a promising direction for future investigation.

The solution for *Step* 2 is computed using a fully implicit finite difference method. Let index j be fixed and solve the following equations using the Thomas algorithm:

$$\frac{\psi_{ij}^{**} - \psi_{ij}^{*}}{\Delta t} = \frac{\psi_{i-1,j}^{**} - 2\psi_{ij}^{**} + \psi_{i+1,j}^{**}}{h^2}, \tag{18}$$

where we use zero Neumann boundary condition

$$\psi_{0j}^{**} = \psi_{1j}^{**} \quad \text{and} \quad \psi_{N_x+1,j}^{**} = \psi_{N_x j}^{**}. \tag{19}$$

In a similar manner, the solution for *Step* 3 is determined through a fully implicit finite difference method. Let index i be fixed and solve the following equations using the Thomas algorithm:

$$\frac{\phi_{ij}^{n+1} - \psi_{ij}^{**}}{\Delta t} = \frac{\phi_{i,j-1}^{n+1} - 2\phi_{ij}^{n+1} + \phi_{i,j+1}^{n+1}}{h^2}, \tag{20}$$

where we use

$$\phi_{i0}^{n+1} = \phi_{i1}^{n+1} \quad \text{and} \quad \phi_{i,N_y+1}^{n+1} = \phi_{iN_y}^{n+1}. \tag{21}$$

The proposed computational scheme unconditionally preserves the maximum principle: if $\|\phi^0\|_\infty \le \phi_\beta$, then $\|\phi^k\|_\infty \le \phi_\beta$ for any $k \ge 1$. More precisely, suppose $\|\phi^n\|_\infty \le \phi_\beta$ for any n, then from Eq. (13), we get

$$|\psi_{ij}^{*}| = \frac{\phi_\beta |\phi_{ij}^n|}{\sqrt{e^{-2\phi_\beta^2 A_{ij}^n \Delta t}(\phi_\beta^2 - (\phi_{ij}^n)^2) + (\phi_{ij}^n)^2}} \le \frac{\phi_\beta |\phi_{ij}^n|}{\sqrt{\left(\phi_{ij}^n\right)^2}} = \phi_\beta. \tag{22}$$

It is well known that the fully implicit finite difference methods for Eqs. (18) and (20) preserve the maximum principle. Therefore, the condition $\|\phi^{n+1}\|_\infty \le \phi_\beta$ holds for any time step size.

The discrete energy functional corresponding to the continuous formulation given in Eq. (2) is defined as follows:

$$\begin{aligned}\mathcal{E}_h(\phi^n) = &\sum_{i=1}^{N_x}\sum_{j=1}^{N_y} \frac{F(\phi_{ij}^n)}{\epsilon^2} h^2 + \frac{1}{2}\sum_{i=1}^{N_x-1}\sum_{j=1}^{N_y}\left(\phi_{i+1,j}^n - \phi_{ij}^n\right)^2 \\ &+ \frac{1}{2}\sum_{i=1}^{N_x}\sum_{j=1}^{N_y-1}\left(\phi_{i,j+1}^n - \phi_{ij}^n\right)^2,\end{aligned} \tag{23}$$

where the zero Neumann boundary conditions are applied.

3. Computational experiments

3.1. Random initial condition

A random initial condition is chosen on $\Omega = (0,1) \times (0,1)$:

$$\phi(\mathbf{x}, 0) = 0.5\text{rand}(x, y), \tag{24}$$

which is shown in Fig. 3(a). We use $N_x = N_y = 100$, $h = 1/N_x$, $\epsilon = 0.5h$, and $\theta = 0.6$. The critical points for this θ value are approximately $\phi_\alpha = -0.9073$ and $\phi_\beta = 0.9073$. Fig. 3(b), (c), and (d) show the evolution of ϕ after 20 iterations for different time step sizes $\Delta t = 0.1h^2$, $\Delta t = h^2$, and $\Delta t = 10h^2$, respectively. This numerical test demonstrates that the proposed algorithm is indeed unconditionally stable.

To numerically verify that the proposed algorithm satisfies both the maximum principle and energy stability under various time step conditions, we conducted simulations using three different time step sizes: $\Delta t = 0.1h^2$, $1.0h^2$, and $10.0h^2$. We examined the evolution of the maximum and minimum values of the computational solution ϕ^n as well as the discrete energy $\mathcal{E}_h(\phi^n)$ over time. Fig. 4 presents these results, where Fig. 4(a) shows the maximum and minimum values of ϕ^n, and Fig. 4(b) displays the temporal behavior of the discrete energy. As shown in Fig. 4(a), the numerical solutions remain within the bounded interval $[-0.9073, 0.9073]$ for all

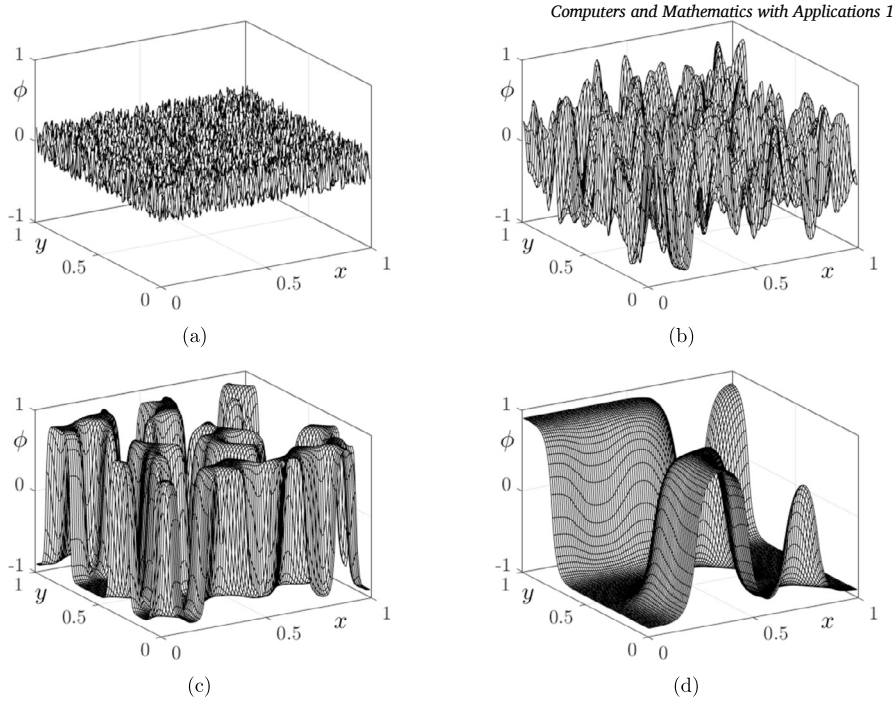


Fig. 3. (a) The initial condition. (b), (c), and (d) represent snapshots of ϕ after 20 iterations, corresponding to time steps $\Delta t = 0.1h^2$, $\Delta t = h^2$, and $\Delta t = 10h^2$, respectively.

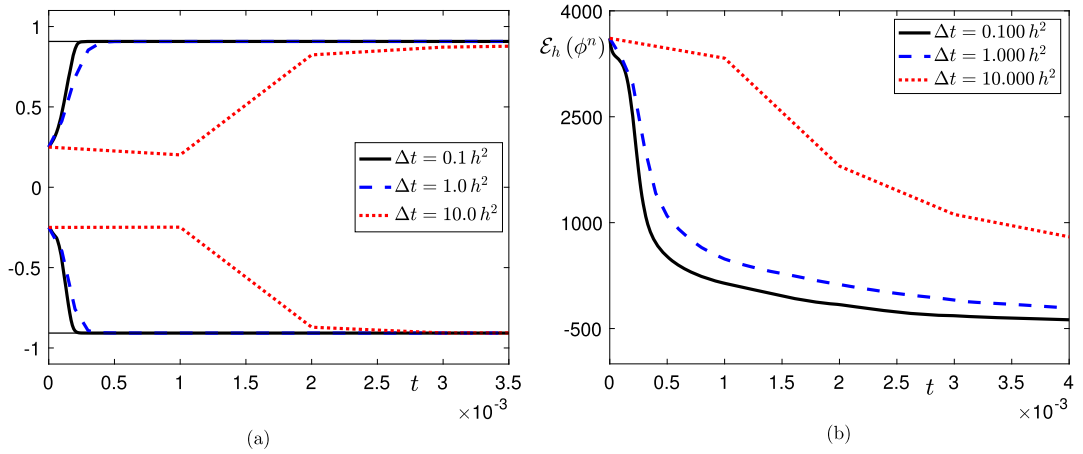


Fig. 4. Time evolution of (a) the maximum and minimum values of the numerical solution ϕ^n for $\Delta t = 0.1h^2,\ 1.0h^2,\ 10.0h^2$, which remain within the critical bounds $\phi_\alpha = -0.9073$ and $\phi_\beta = 0.9073$, and (b) the corresponding discrete energy $\mathcal{E}_h(\phi^n)$, which decreases monotonically over time for all tested time step sizes.

tested time step sizes, which confirms that the algorithm preserves the maximum principle unconditionally. Furthermore, Fig. 4(b) demonstrates that the discrete energy decreases monotonically with time, regardless of the time step size. Although an analytical proof of energy stability has not been provided, these numerical results strongly suggest that the proposed scheme maintains energy stability in practice and ensures reliable temporal evolution.

3.2. Convergence test

To verify the numerical accuracy of the proposed numerical scheme, we conduct convergence tests in both spatial and temporal discretization. An initial condition is used as follows:

$$\phi(\mathbf{x}, 0) = 0.5\cos(\pi x)\cos(\pi y), \tag{25}$$

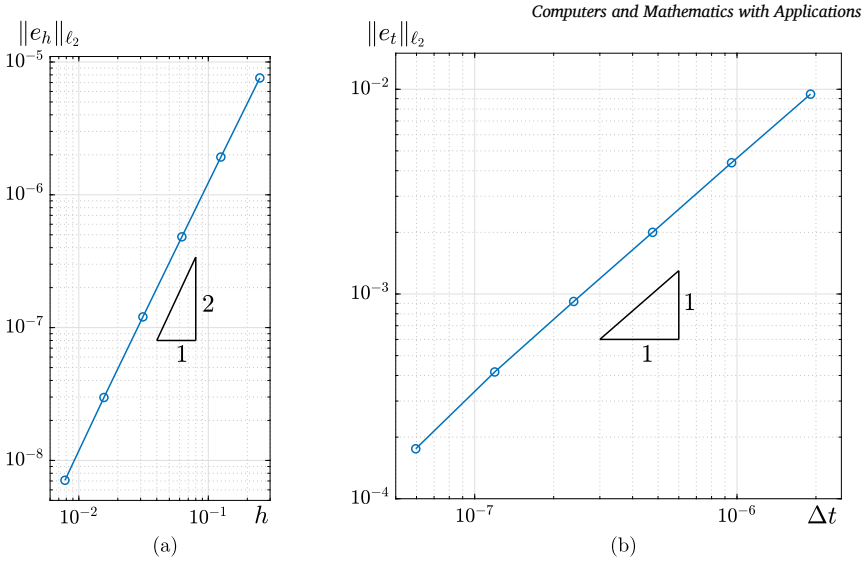


Fig. 5. Convergence errors in (a) space and (b) time are presented for varying spatial step size h and time step size Δt, respectively.

Table 1
ℓ_2-norm discretization errors and corresponding convergence rates with respect to the grid size h.

h	1/4	1/8	1/16	1/32	1/64	1/128
$\|\mathbf{e}_h\|_{\ell_2}$	0.7584e-5	0.1925e-5	0.0483e-5	0.0120e-5	0.0030e-5	0.0007e-5
rate		1.9781	1.9955	2.0028	2.0167	2.0700

Table 2
ℓ_2-norm discretization errors and corresponding convergence rates with respect to the time step size Δt.

Δt	$h_{ref}/2$	$h_{ref}/4$	$h_{ref}/8$	$h_{ref}/16$	$h_{ref}/32$	$h_{ref}/64$
$\|\mathbf{e}_t\|_{\ell_2}$	0.9466e-2	0.4377e-2	0.2001e-2	0.0919e-2	0.0416e-2	0.0175e-2
rate		1.1127	1.1294	1.1223	1.1437	1.2463

on the computational domain $\Omega = (0,1) \times (0,1)$. We use a reference grid size $h_{ref} = 1/512$, reference time step $\Delta t_{ref} = h_{ref}^2/256$, terminal time $T = 8h_{ref}^2$, and $\theta = 0.6$. We first conduct convergence test in spatial discretization for a sequence of grid resolutions $h = 1/2^k$, where $k = 2, 3, \ldots, 7$ with $\Delta t = \Delta t_{ref}$ and $\epsilon = 0.5/4$. Let the discrete ℓ_2-norm error be defined at the terminal time as

$$\|\mathbf{e}\|_{\ell_2} = \sqrt{\frac{1}{N_x N_y} \sum_{i=1}^{N_x} \sum_{j=1}^{N_y} \left(\phi_{ij}^{N_t} - \phi_{\text{ref}}\left(x_i, y_j, N_t \Delta t\right) \right)^2}, \tag{26}$$

where $N_t = T/\Delta t$. Fig. 5(a) depicts the decay of the ℓ_2-error with decreasing mesh size h in a log-log plot, which confirms that the empirical convergence rate is approximately of second order. The numerical error values for each mesh level are listed in Table 1.

Next, we investigate the temporal convergence errors for a sequence of time step size $\Delta = h_{ref}/2^k$, where $k = 1, 2, \ldots, 6$ with $h = h_{ref}$ and $\epsilon = 0.5/4$. Fig. 5(b) demonstrates that the proposed method achieves first-order accuracy in time. The detailed error values are listed in Table 2.

3.3. Effects of θ

Fig. 6 illustrates surface plots with the vertical axis ϕ and horizontal axes x and y, and the temporal evolution under different parameter values of $\theta = 0.4$, 0.8, and 0.95. The figure presents numerical results of the coarsening process at $t = 100\Delta t$, $300\Delta t$, and $1000\Delta t$, labeled as (a), (b), and (c), respectively. Each row corresponds to a specific value of θ, from top to bottom: $\theta = 0.4$, $\theta = 0.8$, and $\theta = 0.95$. The initial condition is the same to Eq. (24) and $\Delta t = 0.1h^2$ is used. The other parameter values remain identical to those used in the previous test. An increase in θ results in a smaller amplitude of ϕ across the domain and wider transition layers between the two critical values during the temporal evolution. The minimum critical values of $F(\phi)$ are $(\phi_\alpha, \phi_\beta) = (-0.9856, 0.9856)$, $(-0.7104, 0.7104)$, and $(-0.3795, 0.3795)$ for $\theta = 0.4$, 0.8, and 0.95, respectively. The temporal evolution highlights the dependence of the solution's behavior on the parameter θ.

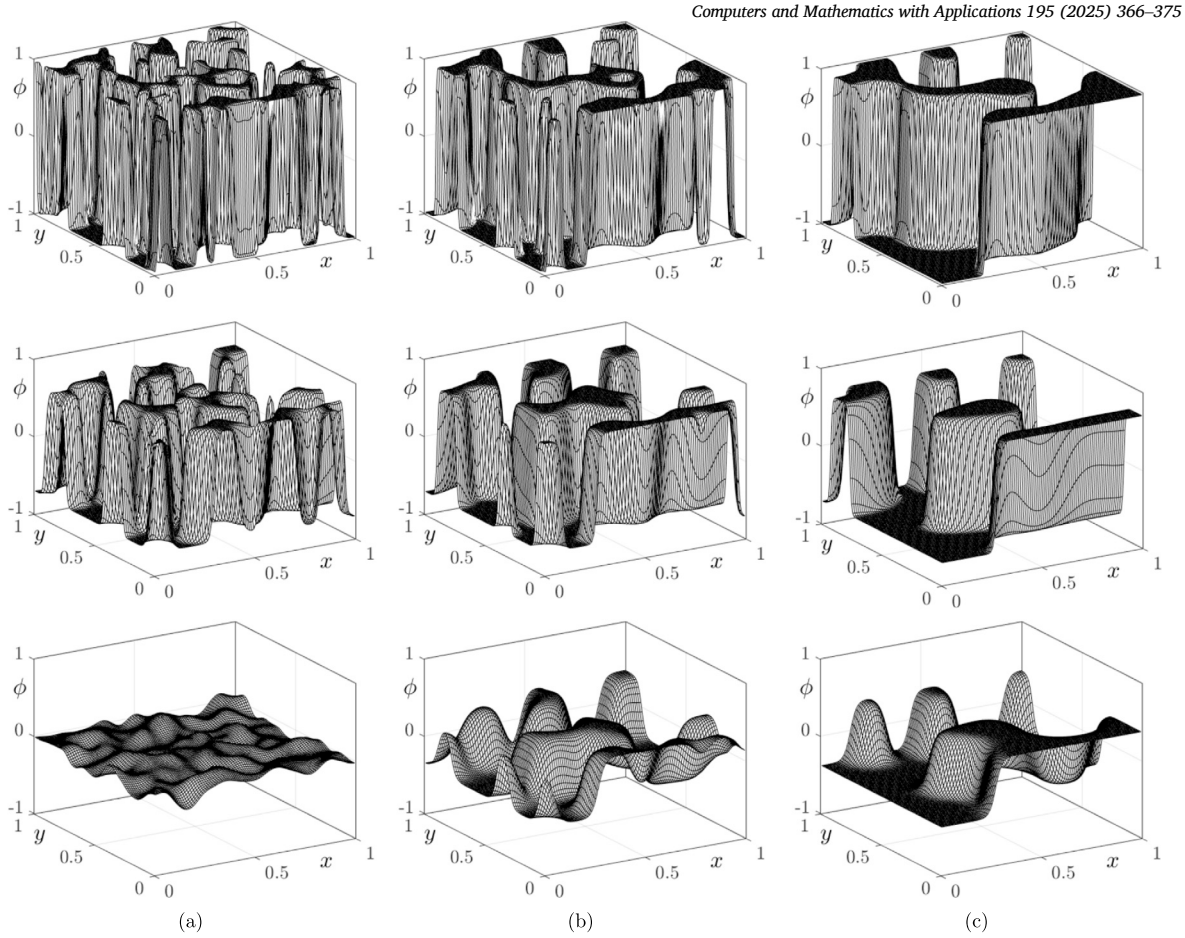


Fig. 6. From top to bottom, the results with $\theta = 0.4,\ 0.8$, and 0.95, respectively. (a), (b), (c) are the time progresses at times $t = 100\Delta t$, $300\Delta t$, and $1000\Delta t$, respectively. Here, $\Delta t = 0.1h^2$ is used.

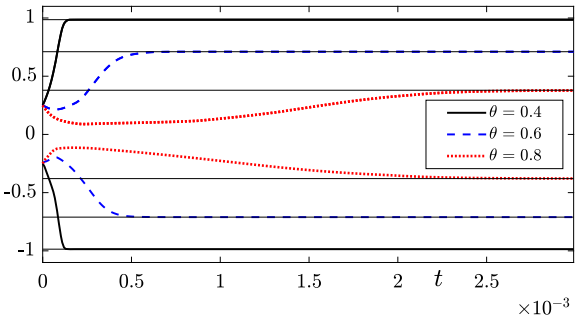


Fig. 7. Time progress of the maximum and minimum of the numerical solutions for $\theta = 0.4,\ 0.8$, and 0.95 with the corresponding critical values of ϕ_α and ϕ_β.

Fig. 7 displays the time progress of the maximum and minimum of the numerical solutions for $\theta = 0.4, 0.8,$ and 0.95, with their corresponding critical values, ϕ_α and ϕ_β. For $\theta = 0.4$ (solid line), the maximum and minimum values of the solution rapidly converge to the critical values, $(\phi_\alpha, \phi_\beta) = (-0.9856, 0.9856)$. For $\theta = 0.6$ (dashed line), the convergence slows compared to $\theta = 0.4$, and takes longer to approach the critical values, $(\phi_\alpha, \phi_\beta) = (-0.7104, 0.7104)$. In the case of $\theta = 0.8$ (dotted line), the convergence is the slowest and the solution values gradually approach $(\phi_\alpha, \phi_\beta) = (-0.3795, 0.3795)$. Larger θ values lead to slower convergence of numerical solutions, which highlights the influence of θ on the temporal dynamics and convergence toward critical values. It can be confirmed that the proposed numerical algorithm satisfies the maximum principle, as the numerical solutions remain bounded by the critical values $(\phi_\alpha, \phi_\beta)$ for all tested values of θ. This demonstrates that the method preserves the inherent stability and boundedness properties of the solution throughout the temporal evolution.

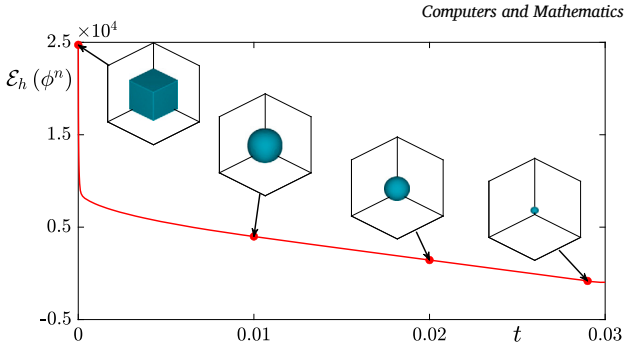


Fig. 8. Time evolution of the discrete energy $\mathcal{E}_h(\phi^n)$ together with snapshots of ϕ at selected time steps. The red curve illustrates the decay of the energy over time, and the insets show the corresponding 3D results of ϕ at each marked time.

3.4. Three-dimensional space

The proposed method can be readily extended to three-dimensional space. To illustrate this, we consider the following example. Let $\Omega = (0,1)\times(0,1)\times(0,1)$ and $\Omega_h = \{(x_i, y_j, z_k) | x_i = h(i-0.5),\ 1 \le i \le N_x, y_j = h(j-0.5),\ 1 \le j \le N_y,\ \text{and}\ z_k = h(k-0.5),\ 1 \le k \le N_z\}$. As in the two-dimensional case, the three-dimensional governing equation is decomposed as follows:

$$Step\ 1: \frac{\partial \phi}{\partial t} = -\frac{F'(\phi)}{\epsilon^2}. \tag{27}$$

$$Step\ 2: \frac{\partial \phi}{\partial t} = \frac{\partial^2 \phi}{\partial x^2}. \tag{28}$$

$$Step\ 3: \frac{\partial \phi}{\partial t} = \frac{\partial^2 \phi}{\partial y^2}. \tag{29}$$

$$Step\ 4: \frac{\partial \phi}{\partial t} = \frac{\partial^2 \phi}{\partial z^2}. \tag{30}$$

The numerical solver adopts an approach analogous to the two-dimensional case. To evaluate the energy stability, we conducted numerical experiments using the following cube-shaped initial condition:

$$\phi(\mathbf{x}, 0) = \begin{cases} \phi_\beta, & \text{if } |x-0.5| \le 0.25,\ |y-0.5| \le 0.25,\ |z-0.5| \le 0.25, \\ \phi_\alpha, & \text{otherwise.} \end{cases} \tag{31}$$

On the computational domain $\Omega = (0,1)\times(0,1)\times(0,1)$. We use $h = 0.01$, $\Delta t = 0.1h^2$, $T = 0.03$, $\epsilon = 0.5h$, and $\theta = 0.6$. The critical points for this θ value are approximately $\phi_\alpha = -0.9073$ and $\phi_\beta = 0.9073$. Fig. 8 shows the temporal evolution of the discrete energy $\mathcal{E}_h(\phi^n)$ together with 3D snapshots of the phase field ϕ at selected time steps. The red curve demonstrates a monotonic decay of the energy over time, which indicates that the proposed scheme preserves energy stability throughout the simulation. The insets visually depict the evolution of ϕ driven by mean curvature flow.

4. Conclusions

In this study, a novel algorithm for solving the AC equation with a logarithmic potential was proposed and analyzed. The method combines the operator splitting approach with a frozen coefficient technique to develop a computationally efficient algorithm that rigorously preserves the maximum principle and exhibits unconditional stability. Analytical proofs and numerical tests have validated the robustness of the proposed algorithm in maintaining the solution's physical bounds and ensure the reliability of the results for practical applications. The computational experiments validated the method's performance under various parameter settings, and demonstrated its capability to handle the challenges associated with the strong nonlinearity and singularity of the logarithmic potential. The numerical results confirmed that the proposed approach provides the maximum principle and stable and accurate simulations of phase dynamics, regardless of the time step size. Furthermore, the influence of temperature-related parameters on phase separation dynamics was systematically explored, which highlighted the versatility of the method in modeling complex physical processes. The proposed scheme serves as a reliable and efficient method for solving phase field equations with logarithmic potentials, and provides the way for further advancements in numerical techniques for similar challenging problems. While the proposed algorithm has been developed and validated in 2D settings with preliminary 3D results, future work will focus on fully 3D domains. Such an investigation is expected to enable more comprehensive simulations of realistic interfacial dynamics and phase transitions in 3D space.

Code availability statement

The main code used to generate the findings of this study is openly available in https://github.com/cfdkim88/CMA_Unconditional_LogarithmAC.

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Data availability

Data will be made available on request.

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